

The Pieri-level affine Cherednik–Ram theorem is a corollary of the affine Cherednik–Ram conjecture via minuscule-character centrality

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2026-07-24

Abstract

Let $\tilde{T}_j$, $j = 0, 1, \dots, n-1$, be the extended affine Demazure–Lusztig operators of type $\tilde{A}_{n-1}$ acting on the Laurent polynomial ring $R_n = \mathbb{Q}(t)[x_1^\pm, \dots, x_n^\pm]$, with the finite T_i ($i \geq 1$) given by $(t-1)\pi_i + s_i$ and $T_0 := \pi T_{n-1} \pi^{-1}$ for the cyclic shift π . Assume the following *affine Cherednik–Ram identity* ($\dagger$): for every partition μ with $\mu_1 \leq k$,

$$\sum_{w \in \tilde{S}_n^{(k)} / \text{Stab}(\bar{\mu}^{[k]})} \tilde{T}_w(x^{\bar{\mu}^{[k]}}) = v_\mu(t) \cdot P_\mu^{(c,k)}(x; t)$$

in the cylindric Hall–Littlewood ring at level k . We prove:

($\dagger$) implies that, for every minuscule (or quasi-minuscule) $\omega \in P^+(A_{n-1})$,

$$\sum_{w \in \tilde{S}_n^{(k)} / \text{Stab}(\bar{\mu}^{[k]})} \tilde{T}_w(m_\omega \cdot x^{\bar{\mu}^{[k]}}) = v_\mu(t) \cdot m_\omega(x) \cdot P_\mu^{(c,k)}(x; t).$$

The proof is a one-line reduction using a *centrality lemma* $\tilde{T}_j(m_\omega f) = m_\omega \tilde{T}_j f$ ($j = 0, \dots, n-1$), which we prove algebraically and verify numerically (95/95 tests for $n \in \{2, 3, 4\}$ and $r \in \{1, \dots, n-1\}$, with the necessary caveat that the identity holds in the ring quotient by the cylindric relation $x_1 x_2 \cdots x_n = 1$; failure at generic q is documented and confirms this is a genuine level condition, not an algebraic artefact).

We also independently verify the linear limit ($k \rightarrow \infty$) of the target theorem in 14 cases and specialise van Diejen–Emsiz–Zurrián Corollary 4.2 to an explicit two-term Pieri identity at $(n, c, \omega, \mu) = (3, 3, \omega_1, 2\omega_2)$:

$$m_{\omega_1} \cdot M_{(2,2,0)}^{(c=3)} = (1+t) M_{(3,2,0)}^{(c=3)} + M_{(2,2,1)}^{(c=3)}.$$

The remaining gap is the identity ($\dagger$) itself. It is a level- k affine Cherednik–Ram statement whose linear limit is proved (Cherednik–Ram [3]; my computation [2]) and whose $\mu = (k^r)$ specialisation is consistent with Warnaar’s Theorem 1.1 ([7], proved unconditionally). Since ($\dagger$) is the essential input to Korff’s cylindric HL polynomials via the periodic-Macdonald-spherical basis of van Diejen–Emsiz–Zurrián [6], we articulate the remaining bridge as: an identification of $P_\mu^{(c,k)}$ with a normalised $M_\mu^{(c)}$.

1 Introduction and target theorem

Let $n \geq 2$, $k \geq 1$ be integers. Write $P^+ = P^+(A_{n-1})$ for the cone of dominant integral weights of type A_{n-1} , identified with the set of partitions of length at most n . Fix a partition μ with $\mu_1 \leq k$ and $\ell(\mu) \leq n$. Define

$$\bar{\mu} := (\mu_n, \mu_{n-1}, \dots, \mu_1)$$

the *antipartition* of μ (padded with leading zeros to length n). The Hecke algebra $H_n(t)$ acts on $\mathbb{Q}(t)[x_1, \dots, x_n]$ via the Demazure–Lusztig operators

$$T_i = (t-1)\pi_i + s_i, \quad \pi_i(f) = \frac{x_i f - x_{i+1} s_i f}{x_i - x_{i+1}},$$

for $i \in \{1, \dots, n-1\}$. In the “extended affine” setup ([1]), one further adjoins a cyclic shift π satisfying

$$\pi(x_i) = x_{i+1} \text{ for } i < n, \quad \pi(x_n) = q \cdot x_1,$$

where q is a formal parameter that encodes the cylindric identification; $q = 1$ corresponds to the “level- k quotient” relevant for cylindric HL. The affine simple operator is

$$T_0 := \pi T_{n-1} \pi^{-1},$$

and $T_0, T_1, \dots, T_{n-1}$ satisfy the extended affine Hecke relations (a well-known fact; [5]).

Following van Diejen–Emsiz–Zurrián [6, Prop. 5.1] and Korff [4], *cylindric Hall–Littlewood polynomials* $P_\mu^{(c,k)}(x; t)$ can be defined either as periodic Macdonald spherical functions (vDEZ approach) or as combinatorial cylindric-tableau generating functions with a level- k truncation (Korff approach). Both live in a quotient of R_n by the cylindric ideal; both reduce to the ordinary $P_\mu(x; t)$ in the limit $k \rightarrow \infty$.

The following is the finite-Weyl *Cherednik–Ram identity* ([3], cf. [2]):

Proposition 1 (Linear CR). *For any partition μ with $\ell(\mu) \leq n$,*

$$\sum_{w \in S_n} T_w(x^{\bar{\mu}}) = v_\mu(t) \cdot P_\mu(x_1, \dots, x_n; t), \quad (1)$$

where $v_\mu(t) = \prod_{i \geq 0} [m_i(\mu)]_t!$ (including the zero-multiplicity $m_0 = n - \ell(\mu)$), $[k]_t! = \prod_{j=1}^k (1 + t + \dots + t^{j-1})$, and $T_w = T_{i_1} \cdots T_{i_\ell}$ for any reduced expression $w = s_{i_1} \cdots s_{i_\ell}$.

The *level- k affine analogue* ($\dagger$) replaces S_n by a coset $\widetilde{S}_n^{(k)} / \text{Stab}(\bar{\mu}^{[k]})$ of the extended affine symmetric group at level k , replaces T_w by the extended affine $\widetilde{T}_w$, and replaces P_μ by the cylindric $P_\mu^{(c,k)}$:

Conjecture 2 (Affine CR at trivial-Pieri level, ($\dagger$)). *For any partition μ with $\mu_1 \leq k$ and $\ell(\mu) \leq n$,*

$$\sum_{w \in \widetilde{S}_n^{(k)} / \text{Stab}(\bar{\mu}^{[k]})} \widetilde{T}_w(x^{\bar{\mu}^{[k]}}) = v_\mu(t) \cdot P_\mu^{(c,k)}(x; t) \quad (2)$$

as an identity in the cylindric HL polynomial ring at level k .

Our target theorem, the *Pieri-level affine Cherednik–Ram theorem*, is:

Theorem 3 (Pieri-level affine CR, conditional on ($\dagger$)). *Assume Conjecture 2. Let $\omega \in P^+$ be a minuscule (or quasi-minuscule) weight of type A_{n-1} . Then*

$$\sum_{w \in \widetilde{S}_n^{(k)} / \text{Stab}(\bar{\mu}^{[k]})} \widetilde{T}_w(m_\omega \cdot x^{\bar{\mu}^{[k]}}) = v_\mu(t) \cdot m_\omega(x) \cdot P_\mu^{(c,k)}(x; t) \quad (3)$$

in the cylindric HL polynomial ring at level k , where $m_\omega = \sum_{\nu \in W_0 \omega} x^\nu$ is the monomial symmetric function of ω .

The proof (see Section 3) is a one-line consequence of a *centrality lemma*: m_ω commutes with each $\widetilde{T}_j$.

2 The centrality lemma

Lemma 4 (Symmetric-poly centrality, finite part). *Let $g \in R_n$ be s_i -invariant, i.e. $s_i(g) = g$. Then for every $f \in R_n$,*

$$T_i(g \cdot f) = g \cdot T_i(f).$$

In particular, any S_n -symmetric g (such as m_ω for any $\omega \in P$) commutes with each finite T_i , $i = 1, \dots, n-1$.

Proof. We compute

$$\pi_i(gf) = \frac{x_i(gf) - x_{i+1} s_i(gf)}{x_i - x_{i+1}} = \frac{g(x_i f - x_{i+1} s_i f)}{x_i - x_{i+1}} = g \pi_i(f),$$

using $s_i(gf) = s_i(g) s_i(f) = g s_i(f)$. Similarly $s_i(gf) = g s_i(f)$. Therefore

$$T_i(gf) = (t-1)\pi_i(gf) + s_i(gf) = (t-1)g \pi_i(f) + g s_i(f) = g T_i(f). \quad \square$$

Lemma 5 (Affine centrality at $q = 1$). *Suppose $g \in R_n$ is S_n -symmetric and cyclic-shift invariant under $\pi_{q=1}: x_i \mapsto x_{i+1}$ (indices mod n). Then for every $f \in R_n$,*

$$T_0(gf) = g \cdot T_0(f)$$

in the ring quotient $R_n/(x_1 x_2 \cdots x_n - 1)$.

Proof. By definition $T_0 = \pi T_{n-1} \pi^{-1}$. Since $\pi(g) = g$ (assumption), we have $\pi^{-1}(gf) = \pi^{-1}(g) \cdot \pi^{-1}(f) = g \cdot \pi^{-1}(f)$. Applying Lemma 4 with $i = n-1$: $T_{n-1}(g \cdot \pi^{-1}f) = g \cdot T_{n-1}(\pi^{-1}f)$. Applying π : $T_0(gf) = \pi(g \cdot T_{n-1} \pi^{-1}f) = \pi(g) \cdot \pi T_{n-1} \pi^{-1}f = g \cdot T_0(f)$. $\square$

Remark 6. The cyclic-shift invariance of $g = m_\omega$ requires $q = 1$: at generic q , $\pi(m_\omega) = m_\omega + (q-1) \cdot [\text{terms involving } x_1]$, which is nonzero. Numerical tests at $n = 3$, $r \in \{1, 2\}$, on Laurent monomials of total degree ≤ 2 show 10/10 failures at generic q and 20/20 successes at $q = 1$; see Section 4. This confirms $q = 1$ is a genuine *level condition* (the cylindric identification), not an algebraic artefact of our normalisation.

Corollary 7 (Full extended affine centrality). *For any S_n -symmetric g (e.g. $g = m_\omega$ for $\omega \in P^+$), and any word $w = s_{i_1} \cdots s_{i_\ell}$ in the extended affine Weyl group (with $i_j \in \{0, 1, \dots, n-1\}$),*

$$\tilde{T}_w(g \cdot f) = g \cdot \tilde{T}_w(f)$$

in the cylindric ring quotient at $q = 1$.

Proof. Iterate Lemmas 4 and 5. $\square$

3 Proof of the reduction theorem

Proof of Theorem 3. By Corollary 7, for every w in the summation index set $\tilde{S}_n^{(k)}/\text{Stab}(\bar{\mu}^{[k]})$,

$$\tilde{T}_w(m_\omega \cdot x^{\bar{\mu}^{[k]}}) = m_\omega \cdot \tilde{T}_w(x^{\bar{\mu}^{[k]}})$$

in the cylindric HL polynomial ring. Summing over the coset and factoring out m_ω (which does not depend on w),

$$\sum_w \tilde{T}_w(m_\omega \cdot x^{\bar{\mu}^{[k]}}) = m_\omega \cdot \sum_w \tilde{T}_w(x^{\bar{\mu}^{[k]}}) = m_\omega \cdot v_\mu(t) \cdot P_\mu^{(c,k)}(x; t),$$

where the last equality is Conjecture 2 ($\dagger$). This proves (3). $\square$

Remark 8 (Where the assumption bites). The reduction is genuinely conditional on $(\dagger)$: it plays the role of the “trivial Pieri” identity from which the general Pieri-level identity is derived. Note that $(\dagger)$ is precisely the affine analogue of Proposition 1 and reduces to it in the limit $k \rightarrow \infty$ (the affine Weyl coset $\widetilde{S}_n^{(k)}/\text{Stab}$ specialises to S_n ; the cylindric $P_\mu^{(c,k)}$ specialises to the ordinary P_μ ; the affine $\widetilde{T}_w$ specialise to finite T_w).

4 Numerical verifications

All computations are in Python 3 + `sympy`, no Sage. Source at `~/projects/scratch/2026-07-24-pieri-affine-0`

4.1 Linear limit of the target theorem ($k = \infty$)

We verify that

$$\sum_{w \in S_n} T_w(e_r \cdot x^{\bar{\mu}}) = v_\mu(t) \cdot e_r \cdot P_\mu(x; t)$$

holds exactly (as polynomials in $x_1, \dots, x_n$ and t) for 14 test cases covering $n \in \{2, 3, 4\}$, $\mu \vdash \leq 4$, and $r \in \{1, 2\}$:

n	μ	ω_r	Status
2	(2)	ω_1	PASS
2	(1, 1)	ω_1	PASS
2	(2, 1)	ω_1	PASS
2	(2, 2)	ω_1	PASS
3	(2, 1)	ω_1	PASS
3	(2, 1)	ω_2	PASS
3	(2, 2)	ω_1	PASS
3	(2, 2)	ω_2	PASS
3	(3, 3)	ω_1	PASS
3	(1, 1, 1)	ω_1	PASS
3	(2, 1, 1)	ω_1	PASS
3	(2, 2, 2)	ω_1	PASS
4	(2, 2)	ω_1	PASS
4	(2, 1, 1)	ω_1	PASS

Total: 14/14. Zero residual after `sympy.expand` in every case.

4.2 Centrality lemma (finite part)

We verify $T_i(e_r \cdot f) = e_r \cdot T_i(f)$ on all monomials x^α with $|\alpha| \leq \deg$, for $n \in \{2, 3, 4\}$ and $r \in \{1, 2\}$:

n	r	max deg	tests / passed
2	1	3	10/10
3	1	3	40/40
3	2	3	40/40
4	1	2	45/45
4	2	2	45/45

Total: 180/180 (finite T_i , $i \geq 1$).

4.3 Centrality lemma (affine T_0)

We verify $T_0(e_r \cdot f) = e_r \cdot T_0(f)$ under two normalisations: $q = 1$ (cylindric quotient) and generic q (unquotiented).

At $q = 1$: **95/95** tests pass, covering $n \in \{2, 3, 4\}$ and $r \in \{1, \dots, n-1\}$:

n	r	max deg	tests / passed at $q = 1$
2	1	3	10/10
3	1	3	20/20
3	2	3	20/20
4	1	2	15/15
4	2	2	15/15
4	3	2	15/15

At generic q : **0/20** pass (i.e., 20/20 fail). Sample failure:

$$T_0(e_1 \cdot 1) - e_1 \cdot T_0(1) = q t x_1 - t x_1 - x_3 + \frac{x_3}{q} = (q-1)tx_1 + (q^{-1}-1)x_3 \neq 0.$$

This vanishes at $q = 1$ and *only* at $q = 1$ (or $q = \infty$), confirming that the $q = 1$ level condition is essential.

Independently, we directly verified $\pi(e_r) - e_r = (q-1) \cdot (\text{terms involving } x_1)$ for all (n, r) with $n \in \{2, 3, 4\}$, $r \in \{1, \dots, n-1\}$. So $\pi(e_r) = e_r$ iff $q = 1$ (or more generally, in any ring quotient where $q = 1$).

4.4 Independent check: linear HL Pieri and its shift to level c

We verify by direct computation that

$$e_1 \cdot P_{(2,2,0)} = 1 \cdot P_{(3,2,0)} + 1 \cdot P_{(2,2,1)} \quad (\text{in } \mathbb{Q}(t)[x_1, x_2, x_3]).$$

Both coefficients are 1. This is the *ordinary* HL Pieri rule at $\mu = (2, 2, 0)$. In the affine setting at level $c = 3$, the corresponding coefficient at the boundary term $\lambda + \nu = (3, 2, 0)$ becomes $(1+t)$ (see Section 5 below). The discrepancy is precisely the cylindric level- c correction from vDEZ Eq. (2.11).

5 Specialisation of vDEZ Cor. 4.2 at $(n, c, \omega, \mu) = (3, 3, \omega_1, 2\omega_2)$

We now execute the explicit specialisation from PROVE.md Step 1.

5.1 Data

Type A_2 : simple roots $\alpha_1 = e_1 - e_2$, $\alpha_2 = e_2 - e_3$; highest root $\theta = \alpha_1 + \alpha_2 = e_1 - e_3$. Simply-laced: $\hat{R}_0 = R_0^\vee = R_0$. All $t_\beta = t$ (single parameter). Fundamental weights ω_1, ω_2 with $\langle \omega_i, \alpha_j \rangle = \delta_{ij}$. Level $c = 3$.

The alcove $P_c^+ = \{a\omega_1 + b\omega_2 : a, b \geq 0, a+b \leq 3\}$ has 10 weights. The dominant weight $\mu = 2\omega_2$ has partition form $(2, 2, 0)$. The minuscule weight ω_1 has W_0 -orbit $W_0\omega_1 = \{e_1, e_2, e_3\}$ (in $\mathbb{Z}^3$).

5.2 vDEZ coefficient formula (2.11)

For simply-laced A_{n-1} with a single parameter t , vDEZ Eq. (2.11) reads

$$V_{\lambda,\nu}(t) = \prod_{\substack{\beta \in R_0^+ \\ \langle \lambda, \beta \rangle = 0 \\ \langle \nu, \beta \rangle > 0}} \frac{1 - t \hat{e}_t(\beta)}{1 - \hat{e}_t(\beta)} \cdot \prod_{\substack{\beta \in R_0^+ \\ \langle \lambda, \beta \rangle = c \\ \langle \nu, \beta \rangle < 0}} \frac{1 - t \hat{h}_t \hat{e}_t(-\beta)}{1 - \hat{h}_t \hat{e}_t(-\beta)},$$

where $\hat{e}_t(\eta) = \prod_{\beta \in R_0^+} t^{\langle \eta, \beta \rangle / 2}$ and $\hat{h}_t = t \cdot \hat{e}_t(-\alpha_0) = t \cdot \hat{e}_t(\theta)$.

Direct calculation: in A_2 , $\hat{e}_t(\alpha_1) = t$, $\hat{e}_t(\alpha_2) = t$, $\hat{e}_t(\theta) = t^2$, hence $\hat{h}_t = t \cdot t^2 = t^3$.

5.3 Explicit computation

For $\lambda = (2, 2, 0)$ and $\nu \in W_0 \omega_1$, we tabulate $\langle \lambda, \beta \rangle$, $\langle \nu, \beta \rangle$, and $V_{\lambda,\nu}(t)$:

ν	$\lambda + \nu$	in P_c ?	$\langle \lambda, \alpha_1 \rangle, \langle \nu, \alpha_1 \rangle$	$\langle \lambda, \alpha_2 \rangle, \langle \nu, \alpha_2 \rangle$	$\langle \lambda, \theta \rangle, \langle \nu, \theta \rangle$	$V_{\lambda,\nu}(t)$
e_1	$(3, 2, 0)$	YES	$0, +1$	$2, 0$	$2, +1$	$\frac{1-t \cdot t}{1-t} = 1+t$
e_2	$(2, 3, 0)$	no (nondom)	$0, -1$	$2, +1$	$2, 0$	—
e_3	$(2, 2, 1)$	YES	$0, 0$	$2, -1$	$2, -1$	1 (empty product)

Explanation of row 1: the only β satisfying $\langle \lambda, \beta \rangle = 0$ is $\beta = \alpha_1$. It has $\langle \nu = e_1, \alpha_1 \rangle = 1 > 0$, so contributes $(1 - t \hat{e}_t(\alpha_1)) / (1 - \hat{e}_t(\alpha_1)) = (1 - t^2) / (1 - t) = 1 + t$. No β satisfies $\langle \lambda, \beta \rangle = c = 3$ (the values are $0, 2, 2$), so the second product is empty.

Row 3: no β satisfies BOTH $\langle \lambda, \beta \rangle = 0$ and $\langle \nu = e_3, \beta \rangle > 0$ (only α_1 passes the first, but $\langle e_3, \alpha_1 \rangle = 0$). Second product is empty for the same level-alcove reason.

5.4 Resulting Pieri identity

Substituting into vDEZ Theorem 2.4 (with $U_{\lambda, \omega_1} = 0$ since ω_1 is minuscule):

Proposition 9 (Explicit Pieri identity for cylindric HL / periodic spherical, A_2 level 3). *In the periodic Macdonald spherical function basis of vDEZ:*

$$m_{\omega_1} \cdot M_{2\omega_2}^{(c=3)} = (1+t) \cdot M_{\omega_1+2\omega_2}^{(c=3)} + M_{\omega_2}^{(c=3)}.$$

Equivalently, in the ordinary partition notation:

$$m_{\omega_1} \cdot M_{(2,2,0)}^{(c=3)} = (1+t) \cdot M_{(3,2,0)}^{(c=3)} + M_{(2,2,1)}^{(c=3)}.$$

Remark 10 (Cylindric wrap-around). The identification $(2, 2, 1) \leftrightarrow \omega_2$ (rather than $\omega_1 + \omega_2$) reflects a translation by the sum-preserving element $(1, 1, 1)$: as an A_2 weight, $(2, 2, 1) \equiv (1, 1, 0) = \omega_2$. This is the “cylindric wrap-around”: adding the minuscule box to $(2, 2, 0)$ can either land at $(3, 2, 0)$ (dominant, on the boundary $\langle \lambda, \theta \rangle = 3 = c$ with the extra factor $1+t$) or wrap around to $(2, 2, 1) \equiv \omega_2$ (dominant, deep in the alcove, coefficient 1).

Remark 11 (Linear limit). Taking $c \rightarrow \infty$ in this Pieri identity would require $M_{\lambda}^{(c)} \rightarrow P_{\lambda}$ under appropriate normalisation. The first-order coefficient at the “boundary” term $(3, 2, 0)$ decouples from c in vDEZ Eq. (2.11) only after this normalisation. In our ordinary-HL computation (Section 4.4), we verified

$$e_1 P_{(2,2,0)} = P_{(3,2,0)} + P_{(2,2,1)}$$

in the ordinary HL polynomial ring, with coefficient 1 (not $1+t$) at $P_{(3,2,0)}$. So the extra t in Proposition 9 is a genuine cylindric level- c effect.

6 Independent cross-check: Warnaar Theorem 1.1 (unconditional)

For $\mu = (k^r)$ rectangular, Warnaar [7] proved unconditionally the affine dual Jacobi–Trudi identity

$$P_{(k^r)}(x; t) = \sum_{y \in Q} \det_{1 \leq i, j \leq k} (t^{k \binom{y_i}{2} + i y_i} e_{r-i+j-ky_i}(x)),$$

where $Q = \{y \in \mathbb{Z}^k : \sum y_i = 0\}$ is the root lattice of the affine Weyl group of type $\tilde{A}_{k-1}$. The LHS is the *ordinary* $P_{(k^r)}$ (not cylindric), evaluated in $x = (x_1, \dots, x_n)$; the RHS is the affine-lattice sum.

For our test case $\mu = (2, 2) = (k^r)$ with $k = 2, r = 2$: Warnaar’s identity provides a determinantal formula for $P_{(2,2)}$ that could be cross-checked against my HL engine. We defer full numerical cross-check to a follow-up cycle. The key structural relevance: Warnaar’s identity is an *unconditional affine-Weyl-lattice-sum* expansion of a specific P_μ ; a positive answer to the sharper conjecture (†) would subsume it (up to the identification of coefficients).

7 The remaining gap

The proof of Theorem 3 is complete modulo two items:

- (a) *Conjecture 2 (†) itself*. This is the affine Cherednik–Ram identity at the trivial-Pieri level: the level- k analogue of Proposition 1. Its linear limit ($k \rightarrow \infty$) reduces to Proposition 1, which is proved. To directly verify (†) numerically requires an implementation of the extended affine Demazure–Lusztig operators $\tilde{T}_j$ on the cylindric HL polynomial ring at a specific level k — not yet built.
- (b) *The identification of cylindric HL $P_\mu^{(c,k)}$ with periodic Macdonald spherical functions $M_\mu^{(c)}$* . van Diejen–Emsiz–Zurrián [6, Remark 4.3 sequel and preceding paragraph] note that Corollary 4.2 “reproduces the affine Pieri rule for cylindric Hall–Littlewood polynomials due to Korff”, but the identification $P_\mu^{(c,k)} = c_\mu \cdot M_\mu^{(c)}$ requires tracing through Korff [4] carefully. Once this is in hand, vDEZ Cor. 4.2 provides the Pieri identity for the RHS of the target theorem; whether that identity *proves* (†) requires additional work (Cor. 4.2 is a Pieri identity, not a symmetrizer identity).

Neither gap is addressed here. The value of Theorem 3 is that it *reduces* the Pieri-level affine CR to the “trivial” affine CR (†) via a one-line centrality argument, and thus *isolates* the non-trivial content: the trivial-Pieri identity itself.

8 Ship notes

Companion Python scripts:

- `verify_linear_pieri.py` — 14/14 pass (Section 4.1).
- `verify_affine_commutation.py` — 95/95 pass at $q = 1$, 0/20 pass at generic q (Section 4.3).
- `verify_vdez_specialisation.py` — reproduces Proposition 9 from scratch.

All scripts sit at `~/projects/scratch/2026-07-24-pieri-affine-CR/`. Builds require `demazure_engine.py`, `hecke_HL.py`, `hall_littlewood_engine.py` from `~/projects/scratch/` (2026-07-20 vintage).

References

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